

**Prediction during within- and cross-language phrase comprehension: Evidence
from eye-tracking**

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Abstract

Comprehenders actively predict upcoming words during language processing. When the actual input does not match the prediction, previous studies have reported mixed findings: a disconfirmed prediction may either remain activated or become inhibited. In the present study, we thus examined the (residual) activation of disconfirmed predictions by measuring eye-movements in two visual-world eye-tracking experiments. In a within- and a cross-language experiment, Chinese-English bilinguals heard a classifier-noun phrase (e.g., “a drop of water”) in which the classifier was always presented in Mandarin Chinese, their L1, and selected the picture corresponding to the noun from a set of three displayed on the screen. Across trials, the congruency of the classifier-noun phrase was manipulated: the noun could be congruent (e.g., “a drop of water”) or incongruent (e.g., “a drop of books”) with the classifier. While Experiment 1 (N = 40) used the same language (participant’s L1, Chinese) for the classifier and noun, Experiment 2 (N = 40) used different languages (Chinese for the classifier; but participants’ L2, English, for the noun). The results suggested that disconfirmed predictions remain activated for a period of time and may subsequently become less active, for example via inhibition. Importantly, this processing pattern was comparable across the within-language and the cross-language experiments. Together, these findings suggest a dynamic and language-independent trajectory in the processing of disconfirmed predictions.

(219 words)

Keywords: prediction; language processing; activation; inhibition; classifier congruency; visual world paradigm; eye movements

Introduction

Language comprehension unfolds dynamically over time, continuously updating representations as new input becomes available. Prediction, conceptualized as the pre-activation of lexical representations derived from prior semantic context, is typically engaged during language comprehension (DeLong & Kutas, 2020; Grisoni, Boux & Pulvermüller, 2024). Research on prediction during language comprehension has been accumulating over recent years (e.g., Altmann & Mirković, 2009; Bannon et al., 2024; Foucart et al., 2015; Martin et al., 2018; Chun & Kaan, 2019; Lelonekiewicz et al., 2021; Kim et al., 2023; Ness & Meltzer-Asscher, 2018; Pickering & Gambi, 2018; Rommers & Federmeier, 2018; Ryskin & Nieuwland, 2023; for reports of no reliable prediction effects, please see, Bannon et al., 2025; Kaan & Grüter, 2021). In general, these studies demonstrate that prediction is reliably engaged in highly constraining contexts, in which the semantic context strongly biases the set of possible upcoming words and makes a specific continuation highly predictable.

For example, previous research has established that participants can both recognize (Fischler & Bloom, 1979) and read (Ehrlich & Rayner, 1981) a word more quickly when it is presented in a highly predictable context. Further, electrophysiological studies also found that event-related potentials (ERPs) were influenced by sentence predictability (DeLong & Kutas, 2020; Grisoni, Boux & Pulvermüller, 2024; Grisoni, Miller & Pulvermüller, 2017; Grisoni, Tomasello & Pulvermüller, 2021; Ness & Meltzer-Asscher, 2018). For example, Grisoni, Boux, and Pulvermüller (2024) found that brain activity (defined by the mean ERP amplitude during sentence comprehension) was larger when participants heard a high constraint sentence with a predictable final word than when they heard a low constraint sentence with an unpredictable final word, suggesting that prediction is engaged in a high constraint context.

Next to these studies focusing on prediction in the native language (L1) in a monolingual situation, other studies also aimed to explore whether prediction occurs in a language learned later in life. Some studies have reported that participants do not generate predictions based on prior semantic context during second language (L2)

comprehension, evidenced by the absence of N400 reduction in predictable compared to unpredictable contexts (Ito et al., 2017; Martin et al., 2013). Yet, other studies provided contrary findings, demonstrating that predictions were also made during L2 comprehension (Chun & Kaan, 2019; Dijkgraaf et al. 2017; Foucart, Martin et al., 2014; Foucart, Ruiz-Tada & Costa, 2016). This inconsistent data pattern across studies could be due to a weaker predictive ability in L2 (Martin et al., 2013) or a L2 processing delay (Dijkgraaf et al., 2017; Momenian et al., 2024), suggesting language-specific differences between native and non-native language processing during prediction. However, the failure to observe a prediction effect in the L2 may alternatively result from the use of indices that do not directly capture predictive processing (cf. Ito et al., 2017; Martin et al., 2013). For example, Ito and colleagues (2017) proposed that the N400 effect may reflect an integration process combining top-down contextual information with bottom-up activation, rather than serving as a direct measure of prediction.

Finally, there is evidence that prediction can interact with language switching (also referred to as code-switching). For example, Liao and Chan (2016) explored semantic expectedness and language switching in auditory sentence comprehension in an EEG study. In their study, the last word of a sentence was either presented in the same or in a different language than the rest of the sentence. Additionally, the expectedness of this last word was manipulated by using words that had a high vs. low expectancy in the specific sentence context. The authors observed an interaction between language switching and expectancy – however only in the first 250 to 350 ms after target word onset.

In everyday language comprehension, not all incoming input confirms our prediction. When predicted words are disconfirmed, two possible outcomes may occur. One possibility is that the predicted word is actively inhibited (DeLong & Kutas, 2020; Kim et al., 2023; Kutas, 1993; Ness & Meltzer-Asscher, 2018; Sánchez-Meléndez et al., 2024). For example, a study by Kim et al. (2023) found that recognizing a prediction-matching word ‘knife’ in a lexical decision task was slower after hearing ‘Father carved the turkey with a path’ (i.e., sentence with prediction-violated final

word), compared with hearing ‘Father carved the turkey with a’ (i.e., sentence without final word). This suggests that the prediction-matching word ‘knife’ had to be inhibited to be able to comprehend the sentence ‘Father carved the turkey with a path’.

However, another possibility is that the predicted word remains activated. This possibility has been supported by the study of Rommers and Federmeier (2018). In their study, participants read weakly constraining sentences in which the critical sentence-final word was either a word presented two sentences earlier, a word that had been predicted but subsequently disconfirmed in a highly constraining context two sentences earlier, or an entirely new word that had not been previously encountered. Their results showed that, relative to new words, both previously seen words and previously expected but disconfirmed words elicited a reduced N400, suggesting that disconfirmed predictions remained activated (i.e., lingered) in memory.

The experimental designs used to support the two opposing accounts (i.e., inhibition vs. activation) were similar to some extent, as both involved inducing a prediction in a highly constraining context and subsequently re-presenting the previously predicted word. Consequently, the conclusions drawn from findings based on the re-presenting of a previously predicted but then disconfirmed word were necessarily inferential, rather than based on direct evidence of online inhibition or activation of that word at the moment of disconfirmation. Thus, it remains an open empirical question whether inhibition and activation accounts should be regarded as mutually exclusive explanations, or whether they can coexist, potentially reflecting different temporal stages of processing, a possibility that has not yet been directly examined in previous experimental designs (e.g., Kim et al., 2023; Rommers & Federmeier, 2018). The present study aims to address this issue by examining whether (and for how long) a disconfirmed prediction remains activated in both within-language and cross-language contexts, and whether the level of activation differs between these contexts. If residual activation is observed, this would suggest that the activation account plays a role in processing disconfirmed predictions. In contrast, if no such activation is detected, this could potentially lend support to the inhibition account.

To address this question, the present study adopted the visual world paradigm

(VWP). This paradigm has been extensively validated as a sensitive online measure of lexical processing, capable of indexing the activation strength of lexical representations (e.g., Allopenna et al., 1998; Huettig et al., 2011; Roembke & McMurray, 2016; Roembke & McMurray, 2025). In this paradigm, participants' eye movements are recorded while they view a set of visual objects and listen to a target word. Critically, a robust finding is that participants' visual attention is drawn not only to the target object but also to competitor objects semantically related to the target (e.g., Huettig & Altmann, 2005; Yee & Sedivy, 2006). Such fixation patterns provide evidence for the co-activation of the target noun and its competitors during real-time language comprehension. Accordingly, the VWP offers a suitable method for examining whether the disconfirmed prediction remains activated (at least for some time).

To manipulate predictability, the present study used classifier congruency. Both Chinese and English have a rich system of classifiers or classifier-like expressions (Allan, 1977, 1978) and in both languages, classifier convey semantic and grammatical information about the following noun (cf. Grüter et al., 2020). For example, the classifier “a drop of” provides information about the quantity and type of entity denoted by the noun. It combines naturally with ‘water’ but not with ‘book’. More generally, classifier–noun combinations in both Chinese and English are not arbitrary: certain classifiers are conventionally associated with specific nouns, and certain nouns typically co-occur with particular classifiers (Lehrer, 1986). Thus, for classifier congruency, our primary interest lies in whether a classifier can be meaningfully and naturally associated with a given noun. Accordingly, we refer to the contrast between ‘a drop of water’ and ‘a drop of books’ as semantic classifier congruence versus semantic classifier incongruence. Previous studies have shown better performance in a classifier congruent condition (e.g., ‘a drop of water’) than in a classifier incongruent condition (e.g., ‘a drop of books’), indicating a classifier congruency effect (cf. Tong et al., 2025a, 2025b; see also Huang & Schiller, 2021; Wang et al., 2019; Zhang & Liu, 2009 albeit classifier congruence was operationalized in a different way in these studies). In addition, two eye-tracking studies in within-language comprehension found that participants looked more to the semantically congruent noun after comprehending

the classifier (Huettig et al., 2010; Mitsugi, 2020). This finding clearly indicates that people use prediction during classifier processing (see also, Grüter et al., 2020).

In an own previous study, we also manipulated classifier congruency for predictability and we explored prediction in within- and cross-language contexts during language production (Tong et al., 2025b). Participants were first presented with a classifier auditorily in their L1 (Mandarin Chinese) or their L2 (English) and then a picture was shown, whose name could be congruent or incongruent with the classifier (e.g., ‘a crew of firemen’ vs. ‘a crew of rain’). Chinese-English participants were asked to name the picture in the language indicated by a color frame. Results showed a within-language classifier congruency effect but no cross-language effect. More specifically, after participants heard a classifier, a congruent picture was named faster than a incongruent picture when classifier presentation and naming response were in the same language (e.g., both Mandarin Chinese). However, a classifier congruency effect was not observed when the picture was named in a language other than that of the classifier. That is, a prediction effect was absent when classifier and response language differed, regardless of the language direction (i.e., L1 classifier-L2 response or L2 classifier-L1 response). Although these findings appear to suggest an absence of prediction effect in a cross-language context, this outcome may be attributable to a task-specific factor. Specifically, we speculated that this absence of a prediction effect could be due to prediction being language-specific which results from an integration of different components (i.e., language and concept) into a task set (Frings et al., 2010; Philipp & Koch, 2010; Vandierendonck et al., 2008) in a language production task. Such an integration might be specifically relevant when the word has to be produced in the correct language, so that both the concept and the language play a role for lexical selection. In the present study, to rule out potential effects of lexical selection, a language-comprehension instead of a language-production task was used, where participants were required to generate a language-unspecific prediction. In addition, although previous studies have demonstrated a prediction effect during L2 comprehension (e.g., Chun & Kaan, 2019; Dijkgraaf et al., 2017; Foucart, Martin et al., 2014; Foucart, Ruiz-Tada & Costa, 2016), these studies have primarily focused on

single-language contexts (i.e., L2 only). By contrast, our previous study examined prediction in a cross-language context and found that the language direction of the classifier and the response did not affect the results. Thus, we chose to use only one language direction in the present study. We selected L1 as the classifier language, as larger prediction effects are generally expected in L1 compared to L2 (see Tong et al., 2025a, 2025b).

Current study

This study aims to determine whether a disconfirmed prediction remains activated in within-language and cross-language contexts, and to assess whether the strength and time course of activation differs between these two contexts. To this end, we adopted the VWP to investigate disconfirmed predictions during comprehension. During the experiment, after previewing three pictures on the screen, Chinese-English bilinguals heard a phrase made up of a classifier and a noun. After hearing the phrase, participants were asked to click on the picture corresponding to the noun. Next to the reaction time (RT) and error rate for the target response (i.e., not clicking on the target picture), we also measured eye movements to pictures that were (in-)congruent to the classifier as a measure of in-the-moment processing.

Trials fell into three categories (see Table 1): 1) One picture was congruent to the classifier, whereas two were incongruent. 2) Two pictures were congruent to the classifier, whereas one was not. 3) Additionally, as a control trial, all three pictures were incongruent to the classifier (all trials in the third category were considered filler trials; see Methods for details). In all three categories, any picture could be the target, meaning that the classifier–noun phrase could be either congruent or incongruent. This manipulation prevented participants from forming strong assumptions a priori about which picture they would have to select. In other words, because participants were frequently asked to click on an incongruent picture as a target picture, so that the design of the study does not inherently bias participants toward looking at the congruent pictures. Thus, if greater looking to a congruent picture was still observed as compared to looking at an incongruent picture, this would indicate that participants generated predictions during classifier comprehension.

We do not predict a general difference in responses and eye-movements between trials containing one congruent picture and those containing two congruent pictures on the screen. The critical aspect is that both allow us to compare how a disconfirmed prediction (i.e., a congruent non-target picture) is processed relative to a non-predicted concept (i.e., an incongruent non-target picture) after the noun was presented. Due to this consideration, for eye-movement analyses, we analyzed only the trials containing both a congruent and an incongruent non-target picture. This included trials with one congruent picture in which the target was an incongruent picture, as well as trials with two congruent pictures in which the target was a congruent picture.

In Experiment 1, the classifier and noun language was the same (i.e., Mandarin Chinese), whereas in Experiment 2, the classifier and noun languages were different (the classifier was always presented in the L1, Mandarin Chinese, and the noun was presented in the L2, English). Based on the research by Rommers and Federmeier (2018), which demonstrates the activation of a disconfirmed prediction, together with findings from our earlier studies (Tong et al., 2025a, 2025b) showing a prediction effect arising from classifier comprehension, we have three expectations concerning the outcomes of our two experiments.

Firstly, based on our previous findings showing a prediction effect during classifier comprehension (Tong et al., 2025a, 2025b), we expected that in both experiments participants would click on the target picture faster when the noun was congruent to the classifier than when it was incongruent.

Secondly, we expected that in both experiments, after hearing the target noun, a predicted word might remain activated even though the target word is different from the predicted word(s). Such activation should result in the looking proportion for the congruent non-target picture being higher than that for the incongruent non-target picture.

Finally, given that a previous study (Liao & Chan, 2016) reported an interaction between semantic expectedness and language switching, we expected to observe a difference in RT and looking proportion between the two experiments. For RTs, classifier-noun congruency might interact with experiment type. That is, participants'

picture-clicking performance may differ between the cross-language context (Experiment 2) and the within-language context (Experiment 1), regardless of whether the noun was congruent or incongruent with the classifier. For looking proportion, one might expect an interaction between picture type and experiment type. That is, the difference in looks toward the congruent versus incongruent non-target pictures may differ between Experiment 1 and Experiment 2.

Experiment 1: Within-Language Prediction

Method

Participants

Forty participants studying at RWTH Aachen University took part in this in-lab study (22 females, 17 males, 1 person did not report their gender; 40 right-handed; age: $M = 25.15$ years, $SD = 4.34$ years). Mandarin Chinese was their native language (L1). All participants gave informed consent.

Materials and apparatus

For the experimental trials, 26 classifiers and 78 nouns/pictures were selected as stimulus material by experimenters (see Appendix A for an English version of the Chinese classifiers and nouns).

Each classifier was used in three different sets. In each set, a classifier was always combined with three different pictures and each of these pictures served once as the referent of the noun presented in the classifier–noun phrase, that is, each picture appeared once as the target. For each classifier, one set was a none-congruent picture set in which neither of the three pictures was congruent to the classifier. The other two sets were either one-congruent picture sets or two-congruent-pictures sets (see Table 1 for an overview of the three set type categories). The target could either be congruent or incongruent, with congruence being defined as whether such combinations are commonly used in daily life (e.g., ‘a drop of water’ being congruent vs. ‘a drop of books’ being incongruent). Across all classifiers and sets, the number of one-congruent-picture and two-congruent-pictures sets was the same. However, some classifiers only occurred in one-congruent-picture sets, while others occurred only in two-congruent pictures sets

or in both kinds of sets. This distribution was based on the number of congruent nouns that could be combined with each classifier. For example, ‘a pair of’ can be combined with a large number of nouns in a meaningful and naturally-used way and, thus, occurred twice in a two-congruent-pictures set, whereas ‘a drop of’ has less congruent nouns so that it occurred twice in a one-congruent-picture set.

Table 1

Classifier-picture set types in Experiment 1 and Experiment 2 with one example classifier-noun phrase each

Classifier-picture set	Classifier	Picture 1	Picture 2	Picture 3
One-congruent	a grain of 一粒 yī lì	Congruent rice 米 mǐ	Incongruent furniture 家具 jiā jù	Incongruent newspaper 报纸 bào zhǐ
Two-congruent	a grain of 一粒 yī lì	Congruent wheat 麦子 mài zi	Congruent corn 玉米 yù mǐ	Incongruent donut 甜甜圈 tián tián quān
None-congruent	a grain of 一粒 yī lì	Incongruent clothes 衣服 yī fú	Incongruent cigarettes 烟 yān	Incongruent plate 盘子 pán zi

Each of the 78 nouns/pictures also occurred in three different sets but was never combined with the same classifier or another noun/picture more than once. To ensure that no single noun/picture stood out among the three nouns/pictures per trial, the nouns were either all semantically related (e.g., all being food/drinks) or all (relatively) unrelated. Additionally, the three nouns did not phonologically overlap in Chinese (and in English, which is important for Experiment 2). A separate group of English native speakers were asked to rate English classifier-noun phrases on a five-point scale (1 = very uncommon, 5 = very common). A paired-sample t-test showed that there was a significant difference between congruent and incongruent classifier-noun combinations (1.46 [incongruent] vs. 4.58 [congruent]), $t(169) = -35.00, p < .001$, indicating that the

classifier-noun phrases used as congruent or incongruent phrases in the experiment clearly differed in terms of their congruency rating.

Corresponding to the 78 nouns, we also selected 78 real-life pictures (350×350 pixels, corresponding to approximately 6.17° by 6.17° of visual angle). Fifty-five of them were taken from the LinguaPix database (Krautz & Keuleers, 2022) and the remaining twenty-three pictures were adapted from licensed online images. To make sure the adapted online pictures corresponded accurately to the nouns, we first selected three different pictures for each of the twenty-three nouns. A separate group of twenty native Chinese speakers was subsequently asked to select the best-matching picture from the three options. The pictures that were selected by the majority of the participants were then used as stimulus material.

All Chinese classifiers and nouns were presented auditorily and were generated by a text-to-speech system in Google Cloud to generate realistic-sounding human voices (Format: wav; name: cmn-CN-WaveNet-B; speed rate: normal; sampling rate: 44100 Hz). Despite the varying durations of each classifier audio file ($M = 762$ ms, $SD = 38$ ms), we kept a constant interval of 1500 ms between the onset of the classifier and the onset of the target noun. That is, there was a silent interval of variable duration before the target-noun onset.

The experiment was programmed and executed using PsychoPy (Peirce et al., 2019), version 2021.2.3. Eye movements were recorded at a sample rate of 1000 Hz with an Eyelink 1000 Plus system (SR Research Ltd, Ottawa, Ontario, Canada), with participants' heads stabilized by a desk-mounted chin and forehead rest. Only the right eye of each participant was tracked. The picture-clicking reaction time (RT; measured as the time from the onset of the target-noun to the timestamp that participant clicked the mouse button after moving the cursor to the correct picture) and accuracy after the target-nouns onset was recorded via PsychoPy.

Procedure

Before the experiment, participants were familiarized with all the pictures. For this familiarization, all pictures and picture names were presented in a PowerPoint presentation (i.e., each slide showed one picture and the corresponding picture noun in

Chinese). Then participants sat in front of a 24" TFT LCD screen (BenQ XL2411P) with a resolution of 1920×1080 pixels positioned at a distance of 90 cm from their eyes. After reading the task instructions (in Chinese) displayed on the screen, a nine-point calibration procedure for the eye-tracker was conducted. Participants began with a practice block consisting of 12 trials to ensure comprehension of the task instructions (both classifier and nouns/pictures used in the practice block differed from those in the actual experiment). A drift correction was conducted at the beginning of each trial (both practice and experimental trials).

In each trial (see Figure 1), three pictures appeared on the screen, with either two pictures at the bottom and one on the top, or vice versa (cf., Roembke & McMurray, 2016). The assignment of the pictures to the locations was randomized in each trial using PsychoPy (Peirce et al., 2019). After picture previewing for 2000 ms, a red dot appeared in the center of the screen. Participants were asked to click with the computer mouse on the red dot if they completed the picture previewing (without time limits). Upon clicking on the red dot, it disappeared and a Chinese classifier-noun phrase was presented auditorily while the three pictures still remained on the screen. Participants were asked to click on the picture that was named in the phrase. After clicking, or when participants failed to respond within 5000 ms, the pictures disappeared. Then, the screen turned blank for 1000 ms before the next trial started.

<Insert Figure 1 about here>

Data analysis

RT analysis. Error trials, defined as incorrect picture selections or no response, as well as all trials with response times (RT) beyond ± 3 standard deviations from the mean RT for each participant were excluded from the RT analysis. A paired-samples t-test was conducted to compare trials in which the noun was congruent vs. incongruent to the classifier (i.e., classifier-noun congruency).

Accuracy analysis. Response accuracy was recorded by PsychoPy. Since it was close to ceiling (99.59 % correct responses), it was not subjected to further analysis.

Looking proportions analyses. We used the VWPre package (version 1.2.3; Porretta et al., 2018) for data preprocessing and the clusterperm package (Barr, n.d.) to

analyze the data. The data were binned into 20-ms intervals, and the looking proportion for each picture was defined as the proportion of samples with a gaze position on that specific picture within each interval. Each trial with more than 25% track loss caused by eye blinks was excluded from the analysis. In addition, trials with an incorrect response, or no response were also excluded. In total, 0.41% of all trials were excluded from looking proportion analyses. The time window of interest was defined as the interval from the start of classifier presentation to the average end of the noun presentation. This interval lasted 1500 ms for classifier presentation and 780 ms for noun presentation¹, resulting in a total duration of 2280 ms. As our second and third hypotheses referred to the time window after target noun presentation, only the looking proportions for non-targets were of interest. To evaluate the experimental hypothesis, the analysis focused exclusively on the looking proportions toward the congruent and incongruent non-target pictures. The trials included in the analysis were trials with one congruent picture in which the target was an incongruent picture and trials with two congruent pictures in which the target was a congruent picture, that is, these trials always contained a congruent and an incongruent non-target picture, resulting in equal numbers of non-targets of each type. This also ensures a balanced comparison in which half of the classifiers were paired with a noun semantically congruent with the classifier, whereas the other half were paired with a semantically incongruent noun.

After aggregating the looking proportions across items for each of the two picture types per subject per interval, we performed cluster-based permutation analysis on the looking proportions in the time window of interest. The procedure for conducting the cluster-based permutation analysis was as follows (Barr, 2022): Firstly, at each interval, we ran an ANOVA and extracted the corresponding test statistic (F-value) and p-value. Consecutive intervals with p-values below 0.05 were grouped into clusters, and for each cluster we computed the cluster-mass statistic. Secondly, to generate the null-hypothesis distribution, we randomly permuted picture type labels (i.e., congruent non-

¹ The playback duration of the nouns varied depending on the number of syllables they had. The mean duration of the noun playback was 778 ms (SD = 89 ms). An additional 2 ms was added to make the time-window duration of the two experiments comparable, resulting in a total noun presentation duration of 780 ms for the analysis.

target and incongruent non-target picture) within subjects 1000 times and recomputed the largest cluster-mass statistic for each permuted data set to create the null hypothesis distribution. Finally, each cluster from the original data was compared against this permutation-based null distribution to obtain Monte Carlo p-values.

Results

Reaction time

Trials with errors or extreme RTs (± 3 SDs from individual means) were excluded. Consequently, 1.61% of the total trials were excluded. A paired-samples t-test revealed that participants selected the target object significantly faster in trials where the noun was congruent with the classifier than in trials where the noun was incongruent with the classifier ($t = -6.44$, $p < .001$; 1129 ms vs. 1187 ms), indicating a -classifier congruency effect.

Looking proportions

Cluster-based permutation analysis showed a significant cluster from 620 ms to 1920 ms (cluster mass statistic = 1574.49, $p < 0.01$), showing higher looking proportions for a congruent non-target picture as compared to an incongruent non-target picture (see Figure 2). This significant cluster indicated that the congruent picture attracted greater attention during the classifier presentation period (620–1500 ms), as well as during the early phase of the noun presentation (1500–1920 ms), when it had become irrelevant.

<Insert Figure 2 about here>

Discussion

In Experiment 1, consistent with the findings from our previous studies (Tong et al., 2025a, 2025b), we found a classifier congruency effect for RTs in participants' L1 (Mandarin Chinese). In addition, participants directed more looks toward the congruent picture than the incongruent picture during classifier presentation, indicating that they engaged in semantic prediction during within-language classifier comprehension. This pattern persisted into the early stage of noun presentation even after the congruent picture had become irrelevant, suggesting that the disconfirmed prediction remained

activated for a period of time. Following this residual activation of the disconfirmed prediction during the early stage of noun presentation, no additional significant clusters were observed. This suggests that participants allocated comparable levels of attention to both the congruent and incongruent non-target pictures at that time point.

Experiment 2: Cross-Languages Prediction

Method

Participants

Forty participants studying at RWTH Aachen University took part in this in-lab study (19 females, 19 males, 2 persons did not report their gender; 40 right-handed; age: $M = 23.93$ years, $SD = 3.34$ years). Mandarin Chinese was their native language (L1) and English was their second language (L2). They completed a questionnaire about the age of L2 acquisition (age: $M = 6.88$ years, $SD = 2.44$ years), and their self-rated language skills for English compared to those of a native English speaker (7-point scale: 1 = “quite poor”, 7 = “highly proficient”; writing: $M = 4.50$, $SD = 1.30$; speaking: $M = 4.45$, $SD = 1.28$; reading: $M = 4.95$, $SD = 1.16$; $M = 4.58$, $SD = 1.20$), which indicated that they rated themselves as less-fluent English speakers compared with native English speakers.

Materials and apparatus

The materials and apparatus were identical to those used in Experiment 1 except that the target-nouns were presented in English. The audio files of English nouns were generated by a text-to-speech system in Google Cloud (Format: wav; name: en-GB-WaveNet-B; speed rate: normal; sampling rate: 44100 Hz). We intentionally selected this voice type in the text-to-speech system to ensure it resembled the sound of the Chinese classifier, so that participants had the impression of a coherent classifier-language phrase despite the language switch.

Procedure and design

The procedure and the design were the same as in Experiment 1, with the exception (as mentioned previously) that target-nouns were always presented in English. Classifier presentation remained always in Mandarin Chinese.

Data analysis

RT analysis. We analyzed the RT following the same procedure as in Experiment 1.

Accuracy analysis. Response accuracy was recorded by PsychoPy. Since it was close to ceiling (99.35 %), it was not subjected to further analysis.

Looking proportion analyses. We analyzed the looking proportion following the same procedure as in Experiment 1. Trials with over 25% track loss due to blinks an incorrect response, or no response were excluded, resulting in 0.65% of all trials being omitted from the looking proportion analyses. The time window of interest was defined as the interval between the start of the Chinese classifier playback and the end of the English noun playback. This included 1500 ms for the classifier playback and 780 ms for the noun playback, resulting in a total time window of 0 to 2280 ms.²

Results

Reaction time

Error trials, defined as incorrect picture selections or no response, as well as all trials with RTs beyond ± 3 standard deviations from the mean RT for each participant were excluded from the RT analysis. Consequently, 2.45% of the total trials were excluded. A paired-samples t-test showed that participants performed significantly better when the noun was congruent with the classifier than when it was incongruent ($t = -6.85, p < .001$; 1150 ms vs. 1218 ms), demonstrating a classifier congruency effect in a cross-language situation.

Looking proportions

Cluster-based permutation analysis showed a significant cluster from 660 ms to 1800 ms (cluster mass statistic = 1091.19, $p < 0.01$), showing higher looking proportions for congruent non-target pictures as compared to incongruent non-target pictures (see Figure 3). This significant cluster suggested that the congruent picture received more attention, beginning in the classifier presentation window (660–1500 ms) and extending into the early noun presentation window (1500–1800 ms), when it had

² The playback duration of the nouns varied depending on the number of syllables they had. The mean duration of the noun playback was 777 ms (SD = 111 ms). An additional 3 ms was added to make the time-window duration of the two experiments comparable, resulting in a total noun presentation duration of 780 ms for the analysis.

become irrelevant. A further significant cluster between 2000 ms and 2260 ms revealed a reversed pattern (cluster mass statistic = 210.31, $p = .04$), with participants looking more to the incongruent non-target picture than to the congruent non-target picture (see Figure 3)

<Insert Figure 3 about here>

Discussion

As in Experiment 1, we found a classifier congruency effect in the RTs. Additionally, increased looks to the congruent picture compared to the incongruent picture during classifier presentation suggested that participants were making predictions while processing the classifier. That is, participants generated predictions based on their L1, Mandarin Chinese, for target-nouns presented in their L2, English, consistent with cross-language prediction. Again, as in Experiment 1, this pattern continued during the early phase of noun presentation, suggesting a residual activation of disconfirmed predictions. However, unlike in Experiment 1, the congruent non-target picture subsequently received less attention than the incongruent non-target picture.

Between-experiment comparisons

Method

Data analysis

RT analysis. A mixed two-way ANOVA was performed with between-subjects factor experiment type (Experiment 1 vs. Experiment 2) and within-subjects factor classifier–noun congruency (noun congruent vs. incongruent to the classifier) as independent variables.

Looking proportion analyses. The looking proportions were analyzed using the same procedure as in Experiments 1 and 2, with the exception that the labels in experiment type were randomly assigned to the participants and those in picture type were randomly permuted in each participant in order to generate the null-hypothesis distribution. We included the experiment type labels permutation in order to test the main effect of experiment type and the interaction between experiment type (Experiment 1 vs. Experiment 2) and picture type (i.e., congruent vs. incongruent non-

target pictures), see Barr (2022).

Results

Reaction time

An ANOVA with experiment type and classifier–noun congruency as independent variables revealed a significant main effect of classifier–noun congruency ($F(1,78) = 88.29, p < .001, \eta^2_p = .53$), showing a faster response when the noun was congruent to the classifier than when it was incongruent. There was no other main effect or interaction ($ps > .43$), suggesting that this effect did not differ across experiments.

Looking proportions

Cluster-based permutation analysis showed a significant cluster of experiment type from 2000 ms to 2280 ms (cluster mass statistic = 143.38, $p = 0.04$), showing higher looking proportions in Experiment 2 as in Experiment 1. In addition, a significant cluster of picture type from 620 ms to 1900 ms was found (cluster mass statistic = 2630.29, $p < .001$), showing higher looking proportions for congruent non-target pictures as compared to incongruent non-target pictures. However, the interaction between experiment type and picture type did not yield any significant clusters (cluster mass statistic = 107.45, $p = .06$).

General Discussion

The present study investigated whether a disconfirmed prediction remains activated in both within-language and cross-language contexts, and whether the level of activation differs between these contexts. While Experiment 1 in the present study was conducted in a within-language context, Experiment 2 was conducted in a cross-language context. Both experiments employed the visual world paradigm, which allowed us to investigate real-time eye movements following the presentation of a classifier and a noun.

For the RT data in both experiments, we expected a classifier congruency effect, such that participants would click on the picture faster when the noun was congruent with the classifier than when it was incongruent. This hypothesis was confirmed in both experiments, which is consistent with findings from previous studies (Tong et al., 2025a,

2025b). The observed classifier congruency effect indicates the presence of a prediction effect, both within- and across-language contexts. With respect to looking proportions, we focused exclusively on the congruent and incongruent non-target pictures, neither of which matched the presented target noun. Given the design of our experiments, specifically the possibility that participants could click on any picture on the screen, the lexical representations corresponding to both the congruent and incongruent non-target pictures are expected to exhibit comparable levels of activation *a priori*. However, we still found that more looks were directed toward a congruent picture than an incongruent picture in both experiments during the classifier presentation (620 ms – 1500 ms for Experiment 1; 660 ms – 1500 ms for Experiment 2). Consistent with previous eye-tracking studies (Huettig et al., 2010; Mitsugi, 2020), this finding suggests that hearing a classifier activates corresponding concepts. That is, participants formed a prediction about the upcoming word rather than merely activating the lexical representations associated with all pictures displayed on the screen. This is an important confirmation of predicted word activation during comprehension, consistent with previous studies (e.g., Grisoni, Boux & Pulvermüller, 2024; Grisoni, Tomasello & Pulvermüller, 2021).

After the noun was presented, both the congruent and incongruent non-target pictures became irrelevant. In Experiment 1, at the early stage of the noun presentation window, higher looking proportions were still found for the congruent non-target picture as compared to the incongruent picture (1500 ms – 1920 ms). This result suggested that the predicted word was highly activated during classifier comprehension and still remains accessible in memory after hearing the noun. By the end of the noun presentation window, looking proportions were comparable for the two non-target pictures in Experiment 1, suggesting that the disconfirmed prediction is not persistently highly activated. Therefore, only the effects observed in the early stage of the noun presentation window align with the findings of Rommers and Federmeier (2018), whereas the effects in the later stage do not. One plausible explanation for this pattern is that, although the disconfirmed prediction may remain briefly activated, inhibitory processes are likely engaged, actively suppressing residual activation of the prediction during disconfirmation. As a result, the disconfirmed prediction may have been

inhibited down to the same level as an irrelevant and non-predicted concept (i.e., the incongruent non-target picture).

In Experiment 2, during the early noun presentation window (1500 ms – 1800 ms), the congruent non-target picture attracted more looks than the incongruent picture, consistent with the pattern found in Experiment 1. However, during a late noun presentation window (2000 ms – 2260 ms), results showed a reversed data pattern, with lower looking proportions in the congruent non-target picture as compared to the incongruent non-target picture. This reversed pattern lends additional support to the proposed inhibitory mechanism. Specifically, following disconfirmation, the residual activation of the predicted (but ultimately incorrect) concept appears to have been actively suppressed, making looking proportions being even lower than the activation associated with an irrelevant, non-predicted alternative (i.e., the incongruent non-target picture). We suspect that in a cross-language context, a congruent non-target concept interfered more strongly with the target concept than an incongruent non-target concept did.

Taken the results from Experiments 1 and 2 together, our findings provide clear evidence that a predicted but subsequently disconfirmed word is activated for a period of time but that such activation does not persist throughout subsequent processing stages. This pattern at the later stage could potentially reflect the operation of inhibitory processes that serve to suppress the pre-activated (but incorrect) representation following disconfirmation (Kim et al., 2023; Ness & Meltzer-Asscher, 2018; Sánchez-Meléndez et al., 2024). Such inhibition may resolve competition between the disconfirmed prediction and the actual input, ultimately facilitating accurate comprehension. The observed pattern of residual activation followed by the suppression for disconfirmed prediction represents a novel contribution to the literature, as it provides valuable additional detail on the time-varying nature of post-disconfirmation processing in language comprehension.

Regarding the comparison of the two experiments, for RTs, we did not find a main effect of experiment type or the interaction of experiment type and classifier–noun congruency. When comparing looking proportions, we only found a main effect of

experiment type with higher looking proportion in Experiment 2 as compared to Experiment 1. Apart from that, no interaction of experiment type and picture type was found. Thus, although the pattern in the noun response window showed differences when analyzing within- and cross-language conditions in isolation, the difference was not substantial enough to become significant during the between-experiment analyses. The between-experiment comparison reveals that the processing of disconfirmed prediction operates in a similar manner across two language contexts. This finding contrasts with the conclusions reported in Liao and Chan (2016) showing the influence of language switching on prediction. However, this apparent difference likely stems from a distinct research focus: the present study investigated the time course of eye movements associated with words that were predicted but not ultimately presented (i.e., disconfirmed predictions), whereas Liao and Chan (2016) examined ERP responses to actual incoming linguistic input. Therefore, in contrast to Liao and Chan (2016), who found an interaction between semantic expectedness and language switching, our results suggest that the residual activation of disconfirmed predictions is not language-specific. Similarly, any subsequent suppression, if present, may also be independent of language context.

Such interpretations are plausible in light of evidence from two additional studies. To elaborate, the first supporting evidence comes from a very recent study (Yin & Pickering, 2026). In their study, the researchers found that when Chinese-English participants generated a prediction for upcoming English target words in highly constraining English contexts, they also increased their fixations on pictures whose Chinese names were homophones of the Chinese translations of those predictable English words. This finding clearly supports a language-non-selective prediction in a two-language context. Thus, the residual activation observed immediately after prediction disconfirmation should be similar in both cross-language and within-language contexts. The second supporting study, by Sánchez-Meléndez and colleagues (2024), suggested that a domain-general inhibitory control mechanism may be recruited during the suppression of disconfirmed prediction. This suggests that, if an inhibitory process operates at a later stage of target noun processing in our current study,

suppressing a predicted but subsequently disconfirmed word may rely on a domain-general mechanism that functions independently of the language relationship between the input and the preceding context.

However, some limitations of the current study also need to be mentioned. First, although we ensured that the nouns within each trial were either all semantically related or all relatively unrelated, thus preventing any single picture from standing out, some differences in visual salience may nevertheless remain, as it is inherently difficult to fully control both visual salience and semantic congruency to the classifier across the three images in each trial. Yet, given that participants had time to prescreen the pictures before the classifier and noun target were presented, we believe these effects to be negligible (Apfelbaum et al., 2021). Second, as our primary focus concerning classifier-noun congruency was on a meaningful and natural association between the classifier and the noun, we cannot distinguish between semantic and grammatical properties of Mandarin Chinese classifiers (cf. Grüter et al., 2020). That is, for the type of Chinese classifiers we used semantic (in)congruence also coincided with grammatical (in)congruence: classifier–noun pairs in the congruent condition were both semantically and grammatically congruent, whereas those in the incongruent condition were semantically and grammatically incongruent (please note that with respect to English grammatical congruence concerning the singular/plural form of nouns, classifiers and nouns were always grammatically congruent to avoid additional influences). In future research it would thus be interesting to disentangle semantic and grammatical influences of Chinese classifiers by including different types of classifiers. Third, as the present study focused exclusively on prediction generated during L1 comprehension, further empirical work is necessary to establish whether disconfirmed predictions generated during L2 comprehension exhibit the same pattern of language-independent activation and subsequent inhibition.

In conclusion, the present study provides evidence that a disconfirmed prediction remains activated for a period of time and may subsequently be subject to inhibition in both within-language and cross-language contexts. These findings suggest a dynamic trajectory in the processing of disconfirmed predictions. Furthermore, this processing

pattern appears to be comparable across within-language and cross-language contexts.

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Appendix A

Classifier-noun pairs used in Experiment 1 and Experiment 2. Congruent words are bolded.

Classifier	Picture 1	Picture 2	Picture 3	Condition
a barrel of 一桶	biscuits 饼干	beef 牛肉	olives 橄榄	none-congruent-picture
	wine 葡萄酒	chocolates 巧克力	ham 火腿	one-congruent-picture
	oil 石油	beer 啤酒	girls 女孩	two-congruent-pictures
a basket of 一篮	birds 鸟	earrings 耳环	cash 现金	none-congruent-picture
	apples 苹果	yogurt 酸奶	pills 药丸	one-congruent-picture
	eggs 鸡蛋	flowers 鲜花	chopsticks 筷子	two-congruent-pictures
a bottle of 一瓶	hay 干草	donuts 甜甜圈	chess 棋	none-congruent-picture
	milk 牛奶	pills 药丸	oranges 橙子	two-congruent-pictures
	perfume 香水	juice 果汁	wood 木头	two-congruent-pictures
a bowl of 一碗	furniture 家具	ropes 绳子	tissues 纸巾	none-congruent-picture
	noodles 面条	firemen 消防员	crayons 蜡笔	one-congruent-picture
	soup 汤	ribbons 丝带	toothpaste 牙膏	one-congruent-picture

a box of	firemen	boys	doctors	none-congruent-picture
一盒	消防员	男孩	医生	
	matches	cereals	bricks	
	火柴	麦片	砖头	two-congruent-pictures
	donuts	crayons	birds	two-congruent-pictures
	甜甜圈	蜡笔	鸟	
a bundle of	juice	rain	water	none-congruent-picture
一捆	果汁	雨	水	
	hay	candies	snow	one-congruent-picture
	干草	糖果	雪	
	newspapers	wood	oil	two-congruent-pictures
	报纸	木头	石油	
a coil of	fish	chocolates	ice	none-congruent-picture
一卷	鱼	巧克力	冰	
	rope	ink	eggs	one-congruent-picture
	绳子	墨水	鸡蛋	
	wire	ribbon	butter	two-congruent-pictures
	电线	丝带	黄油	
a crew of	ham	corns	eggs	none-congruent-picture
一队	火腿	玉米	鸡蛋	
	firemen	cereals	wire	one-congruent-picture
	消防员	麦片	电线	
	doctors	matches	bread	one-congruent-picture
	医生	火柴	面包	
a cube of	hand	noodles	flower	none-congruent-picture
一块	手	面条	鲜花	
	ice	knife	glue	one-congruent-picture
	冰	刀	胶水	
	candy	butter	flour	two-congruent-pictures
	糖果	黄油	面粉	

a cup of	paper	knives	wire	none-congruent-picture
一杯	纸	刀	电线	
	tea	noodles	beef	
	茶	面条	牛肉	one-congruent-picture
	water	hands	earrings	one-congruent-picture
	水	手	耳环	
a dish of	books	bricks	lipsticks	none-congruent-picture
一盘	书	砖头	口红	
	fish	chips	tea	two-congruent-pictures
	鱼	薯条	茶	
	beef	ice-cream	juice	two-congruent-pictures
	牛肉	冰激凌	果汁	
a drop of	soil	bread	toothpaste	none-congruent-picture
一滴	土	面包	牙膏	
	rain	wheat	book	one-congruent-picture
	雨	麦子	书	
	ink	biscuit	chess	one-congruent-picture
	墨水	饼干	棋	
a grain of	clothes	cigarette	plate	none-congruent-picture
一粒	衣服	烟	盘子	
	rice	furniture	newspaper	one-congruent-picture
	米	家具	报纸	
	wheat	corn	donut	two-congruent-pictures
	麦子	玉米	甜甜圈	
a group of	peanuts	beans	cereals	none-congruent-picture
一群	花生	豆子	麦片	
	girls	cash	beans	one-congruent-picture
	女孩	现金	豆子	
	boys	ice	coins	one-congruent-picture
	男孩	冰	硬币	

a handful of 一把	milk 牛奶	flour 面粉	jam 果酱	none-congruent-picture
	peanuts 花生	honey 蜂蜜	beer 啤酒	one-congruent-picture
	beans 豆子	grapes 葡萄	soup 汤	two-congruent-pictures
a jar of 一罐	shoes 鞋	newspaper 报纸	ribbons 丝带	none-congruent-picture
	jam 果酱	clothes 衣服	plates 盘子	one-congruent-picture
	olives 橄榄	lipsticks 口红	stamps 邮票	one-congruent-picture
a nest of 一窝	butter 黄油	oranges 橙子	ice-cream 冰激凌	none-congruent-picture
	birds 鸟	perfume 香水	cheese 奶酪	one-congruent-picture
	ants 蚂蚁	peanuts 花生	shoes 鞋子	one-congruent-picture
a packet of 一包	snow 雪	ink 墨水	girls 女孩	none-congruent-picture
	cigarettes 烟	biscuits 饼干	wine 电线	two-congruent-pictures
	tissues 纸巾	chocolates 巧克力	ants 蚂蚁	two-congruent-pictures
a pair of 一双	chips 薯条	wheat 麦子	honey 蜂蜜	none-congruent-picture
	earrings 耳环	hands 手	soil 土	two-congruent-pictures
	chopsticks 筷子	shoes 鞋	rice 米	two-congruent-pictures

a pile of	soup	oil	wine	none-congruent-picture
一堆	汤	石油	葡萄酒	
	paper	jam	boys	
	纸	果酱	男孩	one-congruent-picture
	bricks	clothes	rain	two-congruent-pictures
	砖头	衣服	雨	
a set of	rice	grapes	yogurt	none-congruent-picture
一套	米	葡萄	酸奶	
	chess	knives	doctors	two-congruent-pictures
	棋	刀	医生	
	furniture	stamps	apples	two-congruent-pictures
	家具	邮票	苹果	
a shovel of	candies	tea	cheese	none-congruent-picture
一铲	糖果	茶	奶酪	
	snow	corns	ropes	one-congruent-picture
	雪	玉米	绳子	
	soil	paper	grapes	one-congruent-picture
	土	纸	葡萄	
a slice of	pill	glue	crayon	none-congruent-picture
一片	药丸	胶水	蜡笔	
	bread	orange	chip	two-congruent-pictures
	面包	橙子	薯条	
	ham	cheese	ice-cream	two-congruent-pictures
	火腿	奶酪	冰激凌	
a spoonful of	matches	stamps	chopsticks	none-congruent-picture
一勺	火柴	邮票	筷子	
	yogurt	tissues	flowers	one-congruent-picture
	酸奶	纸巾	鲜花	
	honey	flour	cigarettes	two-congruent-pictures
	蜂蜜	面粉	烟	

a stack of 一摞	perfume	beer	ants	none-congruent-picture
	香水	啤酒	蚂蚁	
	coins	plates	water	
	硬币	盘子	水	two-congruent-pictures
	cash	books	milk	two-congruent-pictures
	现金	书	牛奶	
a tube of 一管	apples	coins	wood	
	苹果	硬币	木头	none-congruent-picture
	lipstick	fish	hay	one-congruent-picture
	口红	鱼	干草	
	toothpaste	glue	olives	
	牙膏	胶水	橄榄	two-congruent-pictures
